

CivicTrack

Smart Civic Complaint Management System

Citizens submit · Admins assign · Employees resolve

PROJECT REPORT

Capstone Project Report

Software Design · Full-Stack Web Application

TECHNOLOGY STACK	React · Express · MongoDB · Socket.IO · Clerk
FRONTEND HOSTING	Vercel (global CDN)
BACKEND HOSTING	Render (cloud server)
DATABASE	MongoDB Atlas
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01 Executive Summary

A high-level overview of what CivicTrack is, why it was built, and what it achieves.

CivicTrack is a full-stack web application that modernises the way municipalities manage citizen complaints. The platform replaces phone calls, paper forms, and unanswered emails with a single, transparent digital workflow. Citizens report issues online, administrators coordinate responses through a central dashboard, and field employees update progress in real time.

Built as a Software Design capstone project, CivicTrack demonstrates how modern web technology — real-time communication, map-based location tagging, and role-based access control — can be applied to solve an everyday municipal challenge. The result is a working, deployable platform that covers the full complaint lifecycle from first submission to final resolution.

3

User Roles

4

Lifecycle Stages

8

API Endpoints

100%

Real-Time Updates

Mission

Digitise the civic complaint lifecycle — from a resident's first report to a field employee's final resolution — with full transparency, real-time status visibility, and zero paperwork.

02 Problem Statement & Solution

The core challenge this project addresses and the approach taken to solve it.

Municipal complaint handling has long suffered from a set of interconnected problems. Citizens submit complaints through informal channels — phone, email, or in-person visits — and receive little feedback on what happens next. Administrators manage incoming work through spreadsheets or email threads, making it easy for items to be lost or delayed. Field employees often receive assignments verbally or through informal messaging, with no structured system to record progress or outcomes.

Pain Point	Real-World Impact	CivicTrack Solution
No unified submission channel	Citizens use multiple, untracked channels	Single web portal handles all complaints
No status visibility for citizens	Frustration, repeated follow-up calls	Live status shown on citizen dashboard
Manual assignment by administrators	Delays, mis-routing, lost assignments	One-click assignment from admin panel
No structured record of actions taken	Disputes with no supporting evidence	Immutable history log on every complaint
Separate tools for separate teams	Data fragmented across departments	One platform, role-specific views

03 How the System Works

A plain-English walkthrough of the four-stage complaint journey.

Every complaint in CivicTrack moves through four distinct stages. Each stage has a clear owner and triggers automatic notifications so that every party stays informed without any manual follow-up being required.

1 Stage 1: Submit [PENDING]

A citizen logs in, completes a short form — title, category, description, and address — and drops a pin on an interactive map. An optional photo can be attached. The complaint is immediately visible in the administrator's dashboard.

2 Stage 2: Assign [ASSIGNED]

An administrator reviews the incoming complaint and selects the most suitable employee from a searchable list. With a single click the complaint is assigned and the employee receives an instant notification.

3 Stage 3: Work [IN PROGRESS]

The assigned employee opens their task list, marks the complaint as In Progress when work begins, and proceeds to the location. The citizen's view updates automatically to reflect this change.

4 Stage 4: Resolve [RESOLVED]

Once the work is complete, the employee marks the complaint Resolved and optionally adds resolution notes. The citizen sees a final notification and can view the entire history of their complaint from submission to closure.

04 User Roles & Responsibilities

Each user group has a tailored interface showing only what they need.

Citizen	Administrator	Field Employee
• Register a free account and log in securely • Submit complaints with description, category, and photo • Pin the exact location on an interactive map • Track complaint status in real time • View the complete history of every complaint submitted	• View all complaints across the organisation • Filter and search by category, status, or date range • Assign complaints to employees with one click • Create and manage employee accounts • Access analytics dashboard with charts and trends	• View only the complaints assigned to them • Mark a complaint as In Progress when work begins • Add resolution notes to describe the work done • Mark complaints Resolved to complete the lifecycle • Access a personal task summary dashboard

05 Key Features

What differentiates CivicTrack from a simple web form.

Real-Time Status Updates When a complaint status changes — assigned, started, or resolved — the citizen's browser updates instantly without a page refresh. This is delivered via Socket.IO, the same technology used in live messaging applications.	Interactive Map Pinning Citizens drop a pin on a live map to mark the exact location of the problem. Administrators can use this geographic data to route work more efficiently and to identify hotspot areas across the municipality.
Photo Attachments Citizens can photograph the issue — a broken streetlight, pothole, or overflowing bin — and attach it to the complaint. Field employees see the image before arriving, so they know exactly what to expect.	Admin Analytics Dashboard Administrators view complaint volumes over time, status-distribution charts, and resolution trend lines. This makes it straightforward to identify bottlenecks and measure team performance objectively.
Complete Immutable Audit Log Every action on a complaint — assignment, status change, resolution — is permanently appended to a history log. No entry can be overwritten, creating an unbreakable chain of accountability.	Strict Role-Based Access Each user type sees a completely separate version of the platform. Citizens cannot view the admin panel; employees cannot modify complaints outside their assignment. Access is verified on every page and every API call.

06 Technology Stack & Architecture

The components that power CivicTrack, explained without technical jargon.

CivicTrack separates what users see (the frontend) from where data is stored and processed (the backend). The two layers communicate over standard web connections, with real-time updates delivered through a persistent socket channel that runs alongside regular data requests.

Component	Technology	Plain-English Purpose
User Interface	React + Tailwind CSS	Builds every screen users see and interact with
Page Navigation	React Router	Directs users to the correct screen when they click a link
Login & Identity	Clerk	Handles account creation, login, and session security
Web Server	Node.js + Express	Receives requests from the browser and coordinates all work
Database	MongoDB Atlas	Stores every complaint, user record, and history entry
Live Updates	Socket.IO	Pushes status changes to the browser the instant they happen
Maps	React Leaflet	Renders the interactive map for complaint location pinning

Component	Technology	Plain-English Purpose
Charts	Recharts	Draws analytics graphs on the admin metrics dashboard
Frontend Hosting	Vercel	Serves the website to browsers globally via a fast CDN
Backend Hosting	Render	Keeps the server running 24/7 with automatic restarts

07 Security & Data Privacy

How CivicTrack ensures only the right people access the right information.

Security is woven into every layer of CivicTrack rather than added as an afterthought. The platform applies independent checks at the browser, the server, and the database level so that a failure in one layer does not compromise the others.

- +

Secure Authentication

User sessions are managed by Clerk, a specialist identity provider used by thousands of applications in production. Clerk issues short-lived, digitally signed tokens that expire automatically.
- +

Browser-Level Role Checks

Every page verifies the logged-in user's role before rendering. A citizen who navigates to an administrator URL is redirected immediately to their own dashboard without any data being exposed.
- +

Server-Level Role Enforcement

Even if a browser check were bypassed, the server independently validates the user's role on every single API request before accessing the database. No data operation proceeds without a verified, authorised token.
- +

Data Ownership Isolation

Citizens may only retrieve their own complaints. Employees may only update complaints assigned to them. These boundaries are enforced at the API layer and cannot be circumvented through the browser.
- +

Cross-Origin Protection

The server only accepts requests originating from the approved frontend domain. Requests from unknown websites or automated scripts are rejected before any processing occurs.
- +

Immutable Audit Trail

Every change to a complaint — who made it, what it was, and when — is permanently appended to the history log. Entries cannot be modified or deleted, providing a tamper-evident accountability record.

08 Performance & Scalability

How the platform is optimised today and where it can grow tomorrow.

The current deployment handles hundreds of concurrent users comfortably on minimal infrastructure. The architecture also sets clear foundations for scaling to a full city-wide deployment without a complete rebuild.

Optimisation	What It Does	Benefit to Users
Code Splitting	Each page's code is loaded on demand, not all at once	Faster first-page load time
Database Indexing	Four indexes on frequently queried fields	Fast queries even with thousands of complaints

Optimisation	What It Does	Benefit to Users
Geospatial Index	Spatial index on complaint coordinates	Efficient location-based searches and filtering
Socket Rooms	Live events sent only to subscribed clients	Low server overhead for real-time updates
Cloud CDN	Frontend served from Vercel edge network	Low latency for users in any region
Auto-Scaling Backend	Render restarts and scales the server automatically	High availability without manual intervention

09 Project Metrics & Codebase Overview

A factual snapshot of the scale and structure of the CivicTrack codebase.



The table below summarises the main layers of the codebase and the responsibility of each:

Layer	Key Files / Modules	Responsibility
Frontend Pages	12 role-specific pages	Citizen, Admin, and Employee user workflows
Frontend Hooks	6 data-fetching hooks	Loading state, error handling, real-time subscriptions
Frontend API	complaintApi, adminApi, axiosInstance	All HTTP communication with the backend
Backend Routes	complaints.js, admin.js	REST endpoint wiring and request validation
Backend Services	ComplaintService, EmployeeService	Business logic and complaint lifecycle rules
Backend Policies	ComplaintStatusPolicy, AdminCredentialPolicy	Permission checks and valid transition rules
Database Model	Complaint schema (12+ fields, 4 indexes)	Data structure, validation, and persistence
Realtime Layer	socket.js (client and server)	Socket.IO room subscriptions and event emission

10 Future Roadmap

Prioritised improvements that would take CivicTrack from capstone to production.

CivicTrack is built on a foundation intentionally designed to support growth. The following four phases outline the recommended path toward a fully hardened, city-scale platform:

Phase 1 — Security Hardening

→ Move all credentials out of source code and into environment variables

→ Add strict input validation on all API endpoints

→ Authenticate Socket.IO connections using existing Clerk session tokens

→ Enforce ownership checks so employees can only update their own assigned tasks

Phase 2 — Scale & Performance

- Add server-side pagination for all list endpoints
- Move complaint filtering from the browser to the database query layer
- Introduce a Redis adapter for Socket.IO to support multiple server instances
- Replace random complaint IDs with collision-resistant identifiers

Phase 3 — User Experience

- Add email and push notifications for every status change
- Allow citizens to post follow-up comments on their own complaints
- Introduce department-based auto-routing for common complaint categories
- Improve mobile layout and accessibility across all pages

Phase 4 — Testing & Observability

- Add unit tests for all policy and service modules
- Add integration tests covering every API endpoint
- Add end-to-end tests for the full complaint lifecycle
- Add structured logging, error tracking, and a health monitoring dashboard

Closing Note

CivicTrack demonstrates that a focused application of modern web technology can significantly improve the responsiveness and transparency of public services. The platform is a working, deployable proof-of-concept suitable for real-world piloting by a municipality of any size.